

Air Quality Health Index (AQHI) — Edmonton

4.0

Current AQHI at Edmonton Lendrum (18.14 km from venue)
Forecast Sun 5:00 PM at Edmonton Lendrum (18.14 km from venue): **4.3**

Health Risk	AQHI	At-Risk Population	General Population
Moderate Risk	4.3	Consider reducing or rescheduling strenuous activities outdoors if you are experiencing symptoms.	No need to modify your usual outdoor activities unless you experience symptoms such as coughing and throat irritation.

No active Environment Canada weather alerts for the venue.

Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 24.2°C and feels near 26.1°C. Winds are 9 km/h from NW, gusting near 23 km/h.
- The nearest current AQHI is 4 at Edmonton Lendrum, 18.1 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 5 (forecast tonight: 4, tomorrow: 2).
- A gridded AQHI estimate blending official and community sensors (confidence: High, nearest contributing point 6.1 km away) puts the area near 4.
- The nearest air monitoring station (Leduc Sensor, 12.7 km) reports fine particulate matter (PM2.5) at 43.4 µg/m³.
- A nearby community sensor ('ACA_RMA', 7.4 km) reads 45.9 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 37.6 km to the NW, i.e. arriving from the NW.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-08-22) is 20.8 km N of the venue. The modeled back-trajectory passes within 19.2 km of this fire, making smoke transport to the venue plausible based on wind-field modeling.
- The AQHI forecast for Sun 5:00 PM is 4, based on Edmonton Lendrum, 18.1 km from the venue.
- Thunderstorm conditions appear in the hourly forecast beginning around Sun 8:00 PM.
- Overall operational environmental risk is HIGH. Primary concerns are precipitation, thunderstorm.

Recommendations

- Confirm lightning monitoring, shelter, pause and evacuation procedures.
- Prepare drainage, electrical protection and wet-weather controls.
- Environment Canada AQHI guidance — general population: No need to modify your usual outdoor activities unless you experience symptoms such as coughing and throat irritation.
- Environment Canada AQHI guidance — at-risk populations: Consider reducing or rescheduling strenuous activities outdoors if you are experiencing symptoms.

Hazard Snapshot — Overall Risk: High Risk

AIR QUALITY Moderate Risk 4.3 AQHI	HEAT Moderate Risk 29.3 °C apparent/humidex	WIND Low Risk 37.4 km/h peak gust	PRECIPITATION High Risk 3.6 mm/h maximum	THUNDERSTORM High Risk Sun 8:00 PM first forecast signal
AQHI RATE OF CHANGE Moderate Risk 1.0 AQHI change/hour			WIND SHEAR Low Risk 5.7 ° direction diff (10m vs 120m)	

Local Monitoring

Nearest station 4.0 Edmonton Lendrum · 18.14 km	Blended estimate 4.0 confidence High	PM2.5 (station) 43.4 µg/m³ Leduc Sensor · 12.73 km	PM2.5 (community) 45.9 µg/m³ ACA_RMA · 7.4 km
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Prepared by Alberta Capital Airshed for Edmonton Symphony Orchestra outdoor concerts. Beta decision-support product
— for the interactive map, live cameras and hourly outlook, scan the QR code or visit [https://dkevinm.github.io/](https://dkevinm.github.io/eso_sitrep/)

